Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim : Simulate smoothing and sharpening operations on images using frequency domain filters.**	
Experiment 10	**Date: 23/10/2026**	**Enrolment No:92400133050**

Code :

```
import cv2
import numpy as np
import matplotlib.pyplot as plt


# 1. Read image using image path
img = cv2.imread("/content/images (4).jpg", 0)


# Check whether image is loaded successfully
if img is None:
    print("Error: Image not found. Please check the image path.")
else:
    # 2. Fourier Transform
    F = np.fft.fftshift(np.fft.fft2(img))
    M, N = img.shape


    # 3. Create frequency distance matrix
    u, v = np.meshgrid(
        np.arange(N) - N // 2,
        np.arange(M) - M // 2
    )
    D = np.sqrt(u**2 + v**2)
```

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```
# 4. Create Low-Pass and High-Pass Filters
LPF = (D < 30)
HPF = (D >= 30)


# 5. Apply Low-Pass Filter
F_LPF = F * LPF
smooth = np.abs(
   np.fft.ifft2(np.fft.ifftshift(F_LPF))
)


# 6. Apply High-Pass Filter
F_HPF = F * HPF
high_pass = np.abs(
   np.fft.ifft2(np.fft.ifftshift(F_HPF))
)


# 7. Sharpen the original image
sharp = np.clip(
   img.astype(float) + high_pass,
   0, 255
).astype(np.uint8)


# 8. Display Original, Smooth and Sharpened Images
```

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```
plt.figure(figsize=(15, 5))


plt.subplot(1, 3, 1)
plt.imshow(img, cmap='gray')
plt.title('Original Image')
plt.axis('off')

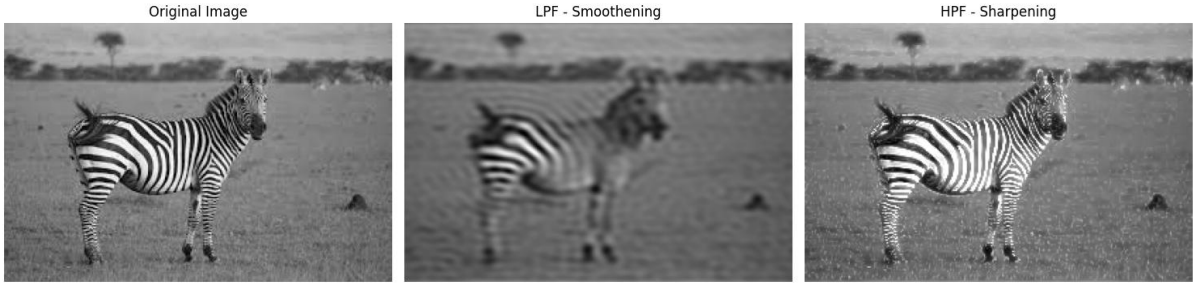

plt.subplot(1, 3, 2)
plt.imshow(smooth, cmap='gray')
plt.title('LPF - Smoothening')
plt.axis('off')


plt.subplot(1, 3, 3)
plt.imshow(sharp, cmap='gray')
plt.title('HPF - Sharpening')
plt.axis('off')


plt.tight_layout()
plt.show()
```

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Output :



Conclusion :

In this experiment, the Fourier Transform was successfully applied to perform image smoothing and sharpening in the frequency domain. The Ideal Low-Pass Filter (LPF) smoothed the image by blocking high frequencies, with the cutoff radius (D) directly controlling the blurring intensity. The High-Pass Filter (HPF) isolated edge information, which, when added to the original image, visibly sharpened features and enhanced contours. The results demonstrate a clear trade-off: smaller cutoff frequencies aggressively remove noise and detail, whereas larger cutoffs preserve more structural information. Ultimately, the experiment validated that manipulating frequency components provides precise, global control over image smoothing and edge enhancement.